

Do World Models Make Better Robots?

A Survey of Evaluation Benchmarks for Predictive Embodied Intelligence

GAYTRI JENA*, UC Berkeley, USA

KAPIL WANASKAR, San Jose State University, USA

VINIJA JAIN, Meta, USA

AMAN CHADHA, Apple, USA

VASU SHARMA, PocketFM, USA

AMITAVA DAS, Pragya Lab, BITS Pilani Goa, India

Robot learning now advances along two tracks that rarely meet. On one side, direct Vision-Language-Action (VLA) policies map observations to actions and are scored by closed-loop task success. On the other, predictive and generative world models forecast future observations and are scored by open-loop prediction or generation quality. A natural question sits between them: does world modelling earn a measurable, closed-loop *advantage* over a direct policy, and for which robotic capabilities? We argue that the field cannot yet answer this question, and that the reason is a gap in how it is measured, not in the models themselves. World-model benchmarks score prediction without ever executing it, while task-success suites host a single policy and never build a world-model versus VLA contrast.

This survey maps the evaluation landscape around that gap. We catalogue 160 web-verified benchmarks spanning 2017 to 2026 and organise them by evaluation mode, robotic capability, and model family into four lanes: policy suites, embodied agents, world-model evaluation, and prediction-to-action bridges. Across the corpus, 138 of 160 benchmarks are model-agnostic and only 11 (7%) build an explicit VLA-versus-world-model contrast; counterfactual capability is almost entirely unmeasured, and only four benchmarks turn prediction into executed action. We contribute an operational taxonomy, a coverage comparison against the eight closest surveys (ours is the only one to cross capability with model family), an evaluation loop that isolates the advantage of prediction, and an actionable protocol of four advantage-aware metrics anchored on named testbeds. The organising claim is not that world models help or do not help, but that answering the question requires benchmarks built to ask it.

CCS Concepts: • **Computing methodologies** → **Robotic planning**; *Computer vision*; *Machine learning*.

Additional Key Words and Phrases: robot learning, world models, evaluation, benchmarks, vision-language-action models, embodied AI, predictive models, survey

1 INTRODUCTION

A robot can be taught to act in two broadly different ways. The first learns a *direct policy*: a function that maps an observation, and often a language instruction, straight to an action. Recent Vision-Language-Action (VLA) models are the prominent instance of this track, and they are judged by running them in an environment and counting task success, on suites such as LIBERO [91], CALVIN [101], RLBench [69], and ManiSkill2 [59]. The second track learns a *world model*: a predictor of future observations that a controller can plan against. This track spans latent-dynamics predictors (DreamerV3, TD-MPC2), action-conditioned video predictors (iVideoGPT, GR-2), and large generative video models (Sora, Cosmos, Genie), and it is judged not by acting but by the quality of what it predicts or generates, on suites such as VBench [68], EvalCrafter [92], and EWMBench [67].

*The authors contributed to this work independently of their roles and employment.

Authors' addresses: Gaytri Jena, UC Berkeley, Berkeley, CA, USA; Kapil Wanaskar, San Jose State University, USA; Vinija Jain, Meta, USA; Aman Chadha, Apple, USA; Vasu Sharma, PocketFM, USA; Amitava Das, Pragya Lab, BITS Pilani Goa, India.

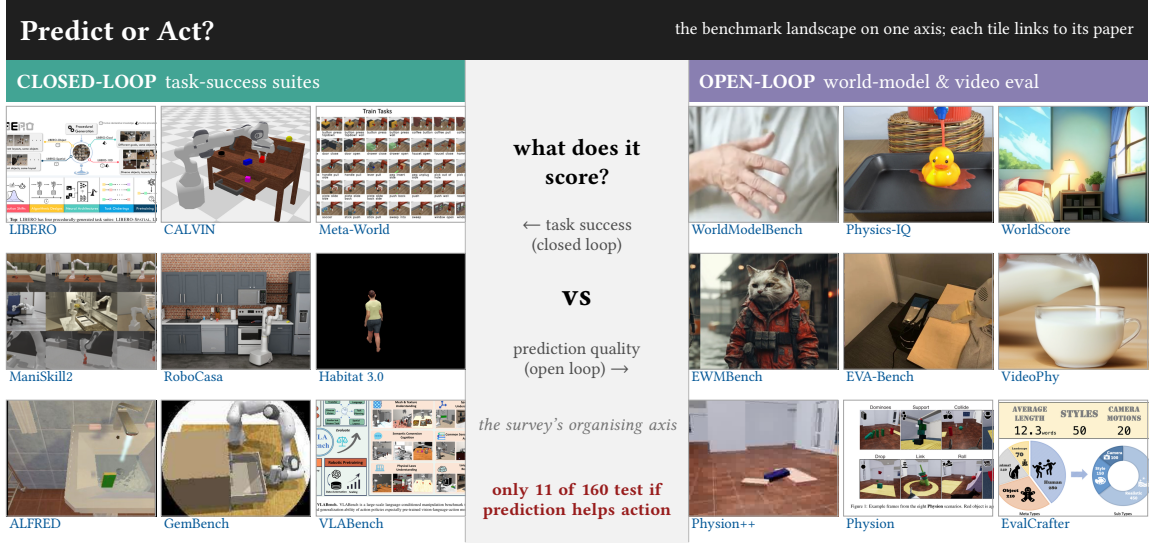


Fig. 1. **Predict or Act?** The survey’s organising question, posed with real benchmarks. *Closed-loop* (left): policy and embodied suites score task success by running a policy in the environment. *Open-loop* (right): world-model and video-generation benchmarks score prediction or generation quality with no action. The centre axis is the question neither pole answers, whether prediction earns a closed-loop advantage, built explicitly by only 11 of 160 benchmarks (Table 10). Every tile is a representative figure from the benchmark’s paper and links to it; per-tile provenance is in Table 13. Images © their respective authors, reproduced for scholarly review.

Between these two tracks sits a question that neither is built to answer: does world modelling earn a measurable, closed-loop *advantage* over a direct policy, and for which robotic capabilities? Figure 1 poses that question with real benchmarks. On the left, closed-loop policy and embodied suites score success by executing a policy in the environment. On the right, open-loop world-model and video-generation suites score prediction or generation with no action taken. The axis down the centre, whether prediction buys a closed-loop advantage, is the one the field has not instrumented.

The gap is methodological rather than empirical. A world-model benchmark scores a rollout for realism or physical plausibility and then stops; it never closes the loop, so it cannot report whether the prediction would have helped a robot act. A task-success suite does close the loop, but it hosts a single policy and is deliberately model-agnostic, so it never contrasts a world-model policy against a direct VLA policy under one protocol. A benchmark that could answer the question must run *both* branches in one closed loop and report the per-capability gain (Section 4, Figure 10). Only a handful of recent “bridge” benchmarks attempt this, and none yet reports a capability-sliced VLA head-to-head.

This survey is therefore about *evaluation*, not about models. It maps the benchmark landscape that surrounds the question and asks which benchmarks could, in principle, establish an advantage of prediction. We catalogue 160 web-verified benchmarks spanning 2017 to 2026 and organise them by evaluation mode, robotic capability, and model family. The headline finding is stark: 138 of the 160 benchmarks are model-agnostic and only 11 (7%) build an explicit VLA-versus-world-model contrast, counterfactual capability is almost entirely unmeasured, and only four benchmarks turn prediction into executed action.

Contributions. This survey offers the following.

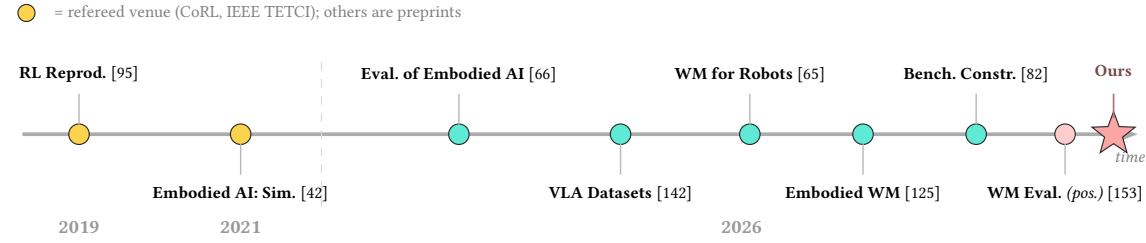


Fig. 2. The closest benchmark/evaluation surveys as a timeline. Two predate the recent wave, an IEEE TETCI survey of embodied-AI simulators and tasks [42] and a CoRL survey of real-robot RL reproducibility [95] (amber = refereed venue); the rest are 2026 preprints on embodied evaluation, VLA data, and world models. None organises the robot-evaluation landscape by evaluation-mode $\times$ capability $\times$ model-family; the sole work reaching the advantage-of-prediction axis is a *position* paper [153]. **Ours** (★) fills that cell; per-axis coverage is in Table 1.

- An **operational taxonomy** of 160 robot-evaluation benchmarks across four evaluation lanes and twelve capability areas (Figure 7), with per-benchmark detail in Tables 3 and 10 and placement rules in Table 2.
- A **profile of the corpus** (Figures 4, 5, 6) and two galleries of the benchmarks’ own subject and results figures (Figures 8, 9), each panel provenance-tracked.
- A **coverage comparison** against the eight closest benchmark and evaluation surveys (Table 1): ours is the only one that crosses robotic capability with model family under a single protocol.
- A **contrast-gap analysis** (Section 4): an evaluation loop that isolates the advantage of prediction (Figure 10), a disambiguation of the overloaded term “world model” (Table 7), and a lane-by-lane account of why the contrast is missing (Table 8).
- An **actionable protocol** for advantage-aware evaluation: four measurable quantities with named testbeds (Table 9, Figure 11).

The organising claim is deliberately modest. We do not argue that world models help, or that they do not. We argue that the question is currently unanswerable with the benchmarks the field has built, and we specify what a benchmark that could answer it would have to do.

1.1 Related surveys and the empty cell

Table 1 places the eight closest surveys and position papers against five axes that a survey of predictive embodied evaluation could organise around: (α) evaluation mode, open- versus closed-loop, as the *primary* axis; (β) robotic capability, such as hidden-state, counterfactual, long-horizon, and social; (γ) a capability-by-model-family cross-tabulation, direct VLA against latent, action-conditioned, and generative world models under one protocol; (δ) a behavioural advantage-of-prediction metric, which we *assess* in the literature rather than adopt; and (ϵ) a comprehensive benchmark catalogue. Existing surveys populate ϵ well and touch α and β in passing, but the capability-by-family cross-tab γ is absent everywhere, and the advantage metric δ is reached only partially, and only by a position paper [153], never by a survey. Figure 2 places these works on a timeline: two predate the recent world-model wave, and the rest cluster in the last two years, none of them crossing capability with family.

Survey	Venue	Yr	α mode	β cap.	γ family	δ metric	ϵ catalog
Evaluation of Embodied AI [66]	Authorea	2026	~	~	✗	✗	✓
VLA Datasets & Bench. [142]	arXiv	2026	✗	✗	✗	✗	✓
Benchmark Construction [82]	arXiv	2026	✗	~	✗	✗	✓
RL Reproducibility [95]	CoRL	2020	✗	✗	✗	✗	✗
World Models for Robots [65]	arXiv	2026	~	✗	~	~	~
Embodied World Models [125]	preprint	2026	~	✗	~	✗	~
WM Evaluation (<i>pos.</i>) [153]	arXiv	2026	✓	~	✗	~	✗
Embodied AI: Simulators [42]	TETCI	2022	~	~	✗	✗	✓
Ours (this survey)	--	2026	✓	✓	✓	✓	✓

Table 1. **Topical coverage of the closest benchmark/evaluation surveys versus ours.** ✓ covered as an organising axis, ~ partial or mentioned in passing, ✗ absent. Axes: α evaluation mode (open- vs closed-loop as the *primary* axis); β capability cuts (hidden-state, counterfactual, long-horizon, social); γ capability $\times$ model-family cross-tab (direct VLA vs latent / action-conditioned / generative world models under one protocol); δ behavioural advantage-of-prediction metric (*assessed*, *not adopted*; e.g. ACTION-ATLAS’s World Advantage Score); ϵ comprehensive benchmark catalogue. This survey is the only work that covers γ , and δ is otherwise reached only *partially*, and only by a *position* paper [153], never by a survey. Venues: CoRL, IEEE TETCI; the remaining rows are arXiv or unrefereed preprints (Authorea, Preprints.org).

2 SURVEY METHODOLOGY AND CORPUS

2.1 Scope, search, and verification

The unit of this survey is the *benchmark*: an individually citeable evaluation artefact, a suite, a dataset paired with a scoring protocol, or an interactive environment with a task distribution, that measures robot, embodied-agent, or world-model behaviour. We deliberately exclude three neighbouring categories that a benchmark survey is often confused with: individual models and policies (the systems a benchmark scores), raw datasets with no scoring protocol, and pure simulators with no task distribution. Where a work contributes both a model and a benchmark, only its benchmark enters the corpus.

Candidates were gathered by parallel web search over arXiv, publisher venues, and project pages, using query strings built from the cross-product of the evaluation lanes (policy, embodied, world-model, bridge) with capability terms (manipulation, navigation, long-horizon, social, counterfactual, physical reasoning, generation quality) and with the terms *benchmark*, *evaluation*, and *suite*. Every candidate was then web-verified against its arXiv or venue page for exact title, first author, year, and identifier; any candidate that could not be verified was dropped rather than recorded from memory, and entries were de-duplicated by both citation key and arXiv identifier. Figure 3 records the resulting funnel in the PRISMA 2020 style [108]; stages that were not logged as counts are marked as such rather than estimated. The procedure yields 160 web-verified benchmarks drawing on 164 references.

2.2 Placement, definitions, and the core-versus-appendix rule

Each benchmark is placed on four axes: its evaluation *lane* (Section 3), the robotic *capability* it primarily stresses, its evaluation *mode* (open-loop prediction or generation quality, closed-loop task success, or a prediction-to-action bridge),

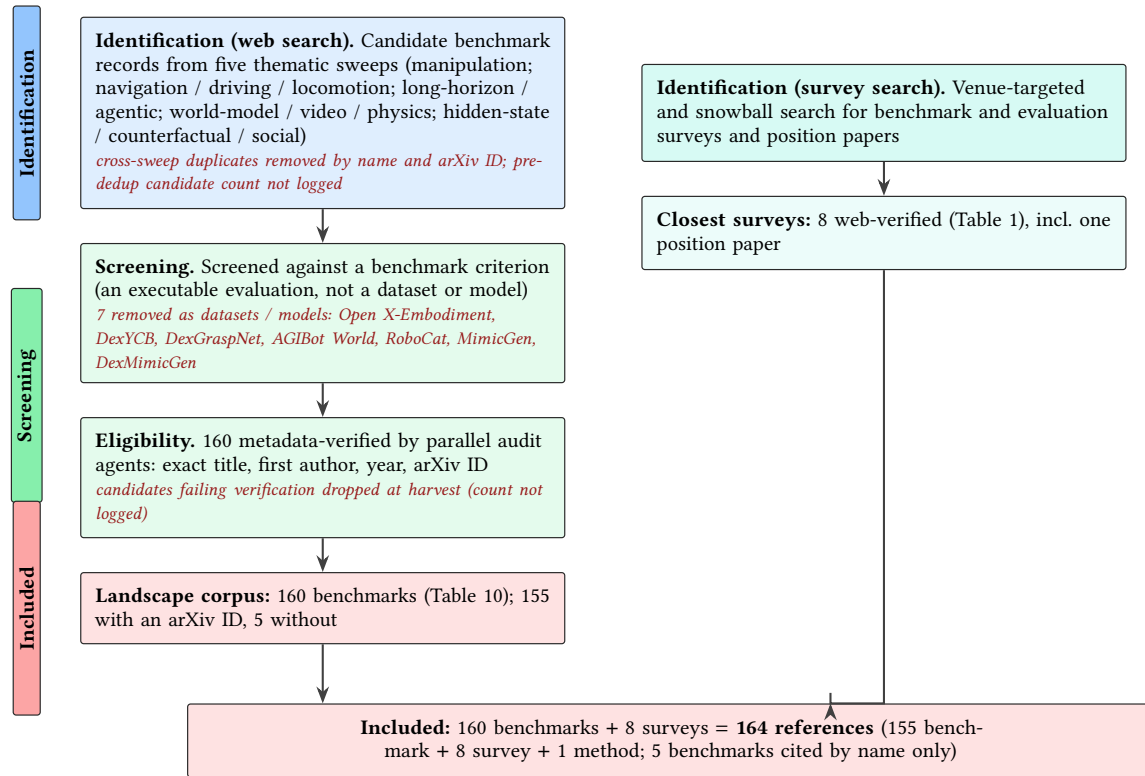


Fig. 3. **Corpus construction** in the PRISMA 2020 style [108]. The left column is the web-search stream behind the landscape corpus (Table 10): five thematic sweeps, de-duplicated by name and arXiv ID, screened to drop datasets and models, and metadata-verified by parallel audit agents. The right column is the venue-targeted and snowball search behind the closest surveys (Table 1). Both streams feed the included set. Candidates whose metadata could not be verified were discarded at harvest; their count was not logged, which we state rather than estimate.

and whether it builds a native VLA-versus-world-model *contrast* (explicit, partial, or model-agnostic). Table 2 states the operational definition used for each placement, so that boundary cases are resolved by a written rule rather than by taste; a benchmark counts as building a contrast, for example, only if a direct-policy and a world-model policy are run under one protocol and compared.

Because the full corpus is large, we split it. A benchmark enters the core taxonomy (Figure 7) when it is cleanly axis-placeable, distinct from its neighbours, and fully characterisable from its paper; the 86 representative benchmarks that meet all three sit in the taxonomy, and the complete set of 160 is catalogued in the landscape table (Table 10, Appendix A). Every count in the survey reconciles with that table.

2.3 The corpus at a glance

Figures 4 and 5 profile the corpus, with all counts re-tallied directly from the catalogue rather than taken from any single paper. The 160 benchmarks divide by lane into 37 policy suites, 85 embodied-agent benchmarks, 34 world-model-evaluation benchmarks, and 4 prediction-to-action bridges, and by capability area into twelve clusters led by long-horizon planning (26), manipulation and dexterity (22), and generation quality (19). By evaluation mode, 113 score

Construct	Counts (a benchmark has it if and only if...)	Does not count
Open-loop eval (open)	it scores prediction or generation <i>quality</i> on fixed inputs without executing the prediction in a feedback loop: video fidelity (FVD), physics or rollout consistency, or frame-level accuracy	any protocol where an action is executed and the environment responds; offline scoring of a policy’s success
Closed-loop eval (closed)	success is measured by running a policy <i>in</i> the environment with state feedback and scoring task outcome: success rate, sub-goal completion, or reward under interaction	scoring a generated video or a static prediction; a metric computed without stepping the environment
Bridge (bridge)	a prediction (imagined rollout, video, or latent plan) is <i>converted into executed actions</i> and scored on the resulting task success	a benchmark that stops at prediction quality; one that never turns a prediction into a control signal
Native VLA-vs-WM contrast (γ)	the benchmark <i>itself</i> runs both a direct Vision-Language-Action policy and a world-model policy under one protocol and reports the head-to-head	a model-agnostic suite that merely <i>hosts</i> either family; a suite evaluated with only one family in the original paper
Advantage-of-prediction (δ)	a metric quantifies the closed-loop <i>gain</i> of a predictive policy over a matched direct-VLA baseline on the same tasks (assessed, not adopted)	absolute success with no matched baseline; open-loop quality scores; a ranking that never isolates the predictive component

Table 2. Operational definitions used to place each benchmark on the evaluation axes (§2). They resolve the ambiguous boundary cases: hosting a policy is not the same as *building* a VLA-vs-world-model contrast (γ), and an absolute success number is not an advantage-of-prediction measurement (δ) without a matched baseline. **open** open-loop, **closed** closed-loop, **bridge** bridge.

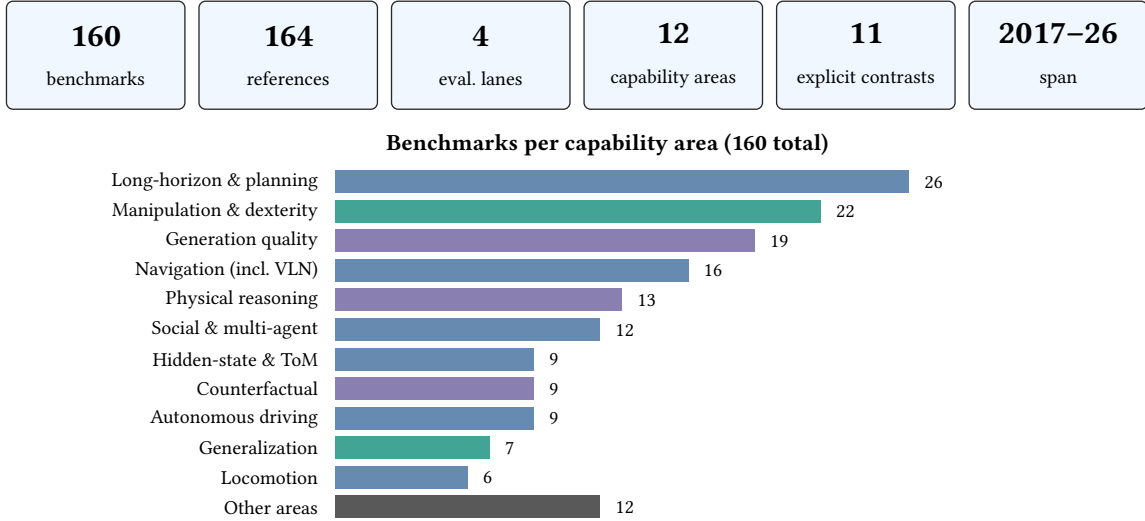


Fig. 4. **The catalogued corpus at a glance.** The survey maps 160 web-verified benchmarks across 4 evaluation lanes, spanning 2017–2026 and totalling 164 references; only 11 build an explicit Vision-Language-Action vs world-model contrast. The bars give benchmarks per capability area, with kindred clusters merged (dexterity into manipulation, vision-language navigation into navigation, theory-of-mind into hidden-state); the twelve areas sum to the 160 works of Table 10. Bars are coloured by the dominant lane of each area (teal policy, blue embodied, purple world-model).

closed-loop task success, 43 score open-loop prediction or generation, and only 4 bridge the two. Publication years show the field’s shape clearly: a steady rise from 2 in 2017 to a peak of 36 in 2024, with the world-model-evaluation lane arriving almost entirely in the last three years (Figure 6). The single most consequential distribution is the contrast axis: 138 of 160 benchmarks are model-agnostic, 11 build a partial contrast, and 11 (7%) build one explicitly. That imbalance is the empirical core of the survey and is taken up in Section 4.

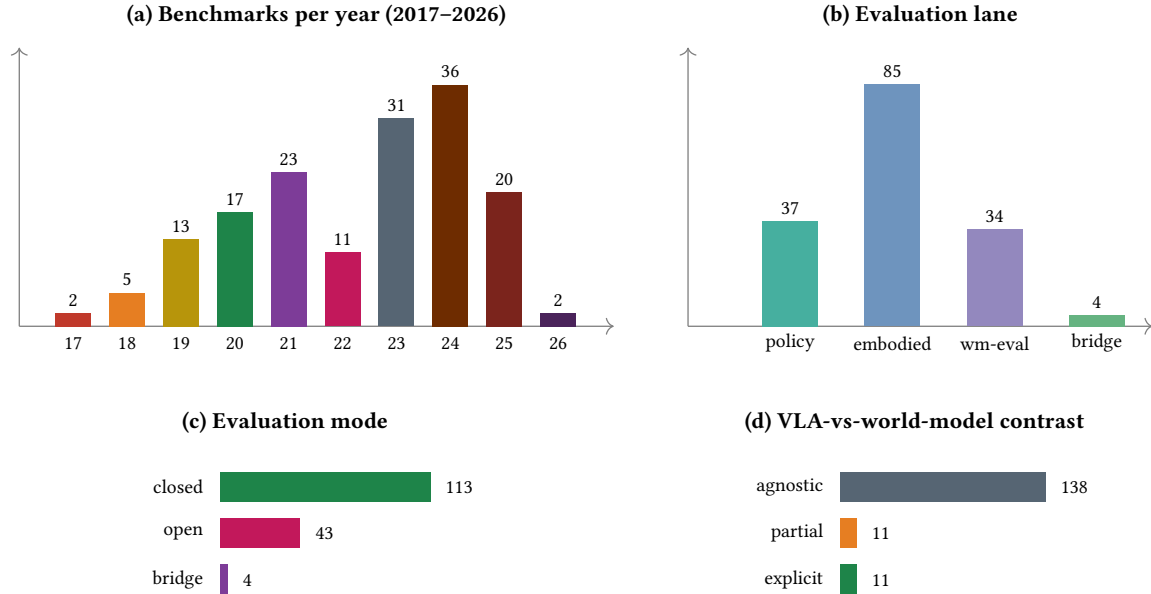


Fig. 5. **Profile of the 160 catalogue benchmarks** (Table 10), re-counted directly from the catalogue. (a) publication year, showing the 2023–2024 surge; (b) evaluation lane (embodied navigation dominates, then policy suites); (c) evaluation mode (closed-loop task success dominates, open-loop prediction is second, only four bridges reach action); (d) whether the benchmark builds a native Vision-Language-Action vs world-model contrast, the survey’s organising question: **138 of 160 are model-agnostic**, and only **11 (7%)** build the contrast explicitly. Counts are the exact column tallies of Table 10.

3 A TAXONOMY OF ROBOT-EVALUATION BENCHMARKS

Figure 7 organises 86 representative benchmarks under a single root and four evaluation lanes. The lanes are ordered by how close their scoring sits to executed behaviour: from closed-loop policy suites (Section 3 §3.1) and embodied-agent benchmarks (§3.2), through open-loop world-model evaluation (§3.3), to the prediction-to-action bridges (§3.4) that alone turn prediction into action. Within each lane, benchmarks are grouped by the capability they stress. Table 3 gives a detailed comparison of a representative subset on year, venue, evaluation mode, and contrast, and Table 4 places one exemplar per lane against a common set of capability dimensions; neither lane’s exemplar builds the VLA-versus-world-model contrast that Section 4 isolates.

Table 3. **Detailed comparison of a representative 31 of the 160 benchmarks** in the landscape (Table 10), grouped by evaluation lane. **Eval:** open open-loop prediction/generation quality, closed closed-loop task success, bridge prediction-to-action bridge. **VLA vs WM:** ✓ native direct-policy vs world-model contrast, - partial, -- model-agnostic. Venues are web-verified; “arXiv” marks preprint-only works.

Benchmark	Year	Venue	Eval	Capability	VLA vs WM
§3.1 Policy / manipulation suites					closed
CALVIN [101]	2021	RA-L	closed	long-horizon language	--
LIBERO [91]	2023	NeurIPS D&B	closed	short-horizon + transfer	--
Meta-World [152]	2020	CoRL	closed	multi-task / meta-RL	--

Table 3 (continued)

Benchmark	Year	Venue	Eval	Capability	VLA vs WM
ManiSkill2 [59]	2023	ICLR	closed	generalizable skills	--
SIMPLER [88]	2024	CoRL	closed	generalization (real vs sim)	--
THE COLOSSEUM [113]	2024	RSS	closed	generalization / robustness	--
RoboCasa [105]	2024	RSS	closed	long-horizon (data scaling)	--
RoboArena [4]	2025	CoRL	closed	real-world generalization	--
GemBench [54]	2024	ICRA	closed	generalization levels	--
VLABench [160]	2025	ICCV	closed	long-horizon reasoning	--
§3.2 Embodied navigation, planning & social					closed
Habitat 3.0 [112]	2023	ICLR	closed	social / multi-agent	--
BEHAVIOR-1K [85]	2024	CoRL	closed	long-horizon household	--
ALFRED [128]	2019	CVPR	closed	long-horizon, partial-obs	--
TEACH [107]	2021	AAAI	closed	hidden-state (dialog)	--
RoboTHOR [38]	2020	CVPR	closed	navigation / sim-to-real	--
SocNavBench [14]	2021	ACM THRI	closed	social navigation	--
LIBERO-Mem [36]	2025	AAAI	closed	hidden-state / occlusion	✓
VIMA-Bench [75]	2023	ICML	closed	multimodal-prompt generalization	--
§3.3 World-model & video-generation evaluation					open
WorldModelBench [86]	2025	arXiv	open	video-WM prediction quality	--
Physics-IQ [103]	2025	WACV	open	physical realism	--
VBench-2.0 [161]	2025	arXiv	open	generation faithfulness	--
WorldScore [41]	2025	ICCV	open	world-gen quality	--
EWMBench [67]	2025	arXiv	open	scene/motion/semantic quality	--
EVA-Bench [33]	2024	ICML	open	embodied video anticipation	--
WorldPrediction [25]	2025	arXiv	open	counterfactual action recognition	--
Physion [12]	2021	NeurIPS D&B	open	physical prediction from vision	--
Physion++ [138]	2023	NeurIPS D&B	open	physical prediction with online property inference	--
§3.4 Bridges (prediction to action)					bridge
WorldSimBench [115]	2024	ICML	bridge	video-to-action consistency	~
World-in-World [157]	2025	ICLR	bridge	closed-loop task success of WMs	~
RoboWM-Bench [73]	2026	arXiv	bridge	prediction executability	~
WorldArena [124]	2026	arXiv	bridge	closed-loop WM arena	~

The two galleries make the corpus concrete. Figure 8 collects the benchmarks’ own *subject* figures, the task and scene images that show what each benchmark depicts, grouped by lane rather than by embodiment. Figure 9 collects the *results* figures, the leaderboards, radar plots, and distributions that show what each benchmark reports. Every panel links to its source paper and is provenance-tracked (Tables 11, 12); benchmarks with no locatable figure are omitted, not fabricated.

3.1 Policy suites

The policy lane (37 benchmarks) scores a policy by executing it and counting task success. Its sub-families run from single- and dual-arm *manipulation* (§3.1.1: ARNOLD [56], PerAct [127], FMB [94], ManiSkill2 [59]), through *generalization*

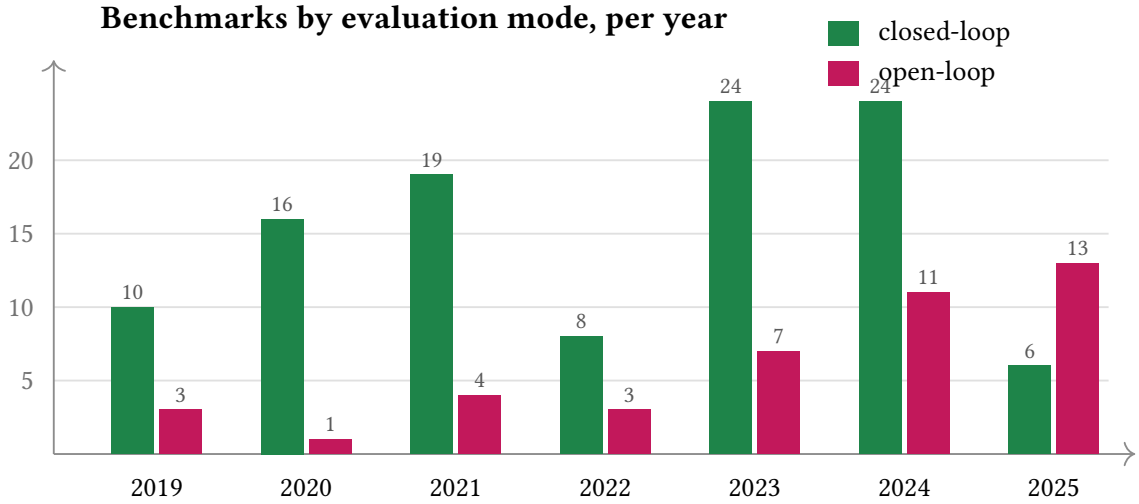


Fig. 6. **The world-model-evaluation wave is recent.** Per-year counts of closed-loop task-success benchmarks (green) and open-loop world-model / video-generation benchmarks (magenta) in the catalogue. Closed-loop suites dominate through 2024; in **2025 open-loop evaluation overtakes them** (13 vs 6) as the world-model-generation literature arrives. The four prediction-to-action bridges are newer still, all appearing in 2024 or later. Years are the benchmarks’ first-release years (2026 is partial and omitted); counts are the tallies of Table 10.

Dimension	LIBERO [91]	Habitat 3.0 [112]	WorldModel-Bench [86]	World-in-World [157]
	<i>policy suite</i>	<i>embodied nav</i>	<i>world-model eval</i>	<i>bridge</i>
What it scores	task success	social-nav success	video-pred. quality	closed-loop WM success
Evaluation mode	closed	closed	open	bridge
Executes actions in a loop?	✓	✓	✗	✓
Native VLA-vs-WM contrast?	✗	✗	✗	~
Capability stressed	short-horizon + transfer	social / multi-agent	physical realism	plan executability
Per-capability slicing?	~	~	✗	✗
Advantage-of-prediction metric?	✗	✗	✗	~
Real / sim / video	sim	sim	video	sim
Counterfactual probing?	✗	✗	✗	✗
Reusable leaderboard?	✓	✓	✓	~

Table 4. Capability matrix, one representative benchmark per evaluation lane. No lane’s exemplar builds a native VLA-vs-world-model contrast: policy and embodied suites are model-agnostic (they *host* any policy), world-model suites score generation quality with no action, and the bridge exemplar reaches closed-loop success but reports only a partial contrast (“visual quality is not task success”). Counterfactual probing is absent throughout. ✓ yes, ~ partial, ✗ no, -- n/a; *open* open-loop, *closed* closed-loop, *bridge* bridge.

suites that vary objects, scenes, and sim-to-real gaps (§3.1.2: LIBERO [91], THE COLOSSEUM [113], SIMPLER [88], VIMA-Bench [75]), to *dexterous* hands (§3.1.4: Bi-DexHands [28], DexArt [9], UniDexGrasp [146]). These suites are the backbone of policy evaluation, and they are deliberately model-agnostic: they host whatever policy a user brings and are careful not to privilege one architecture. That neutrality is exactly why they cannot, on their own, answer whether

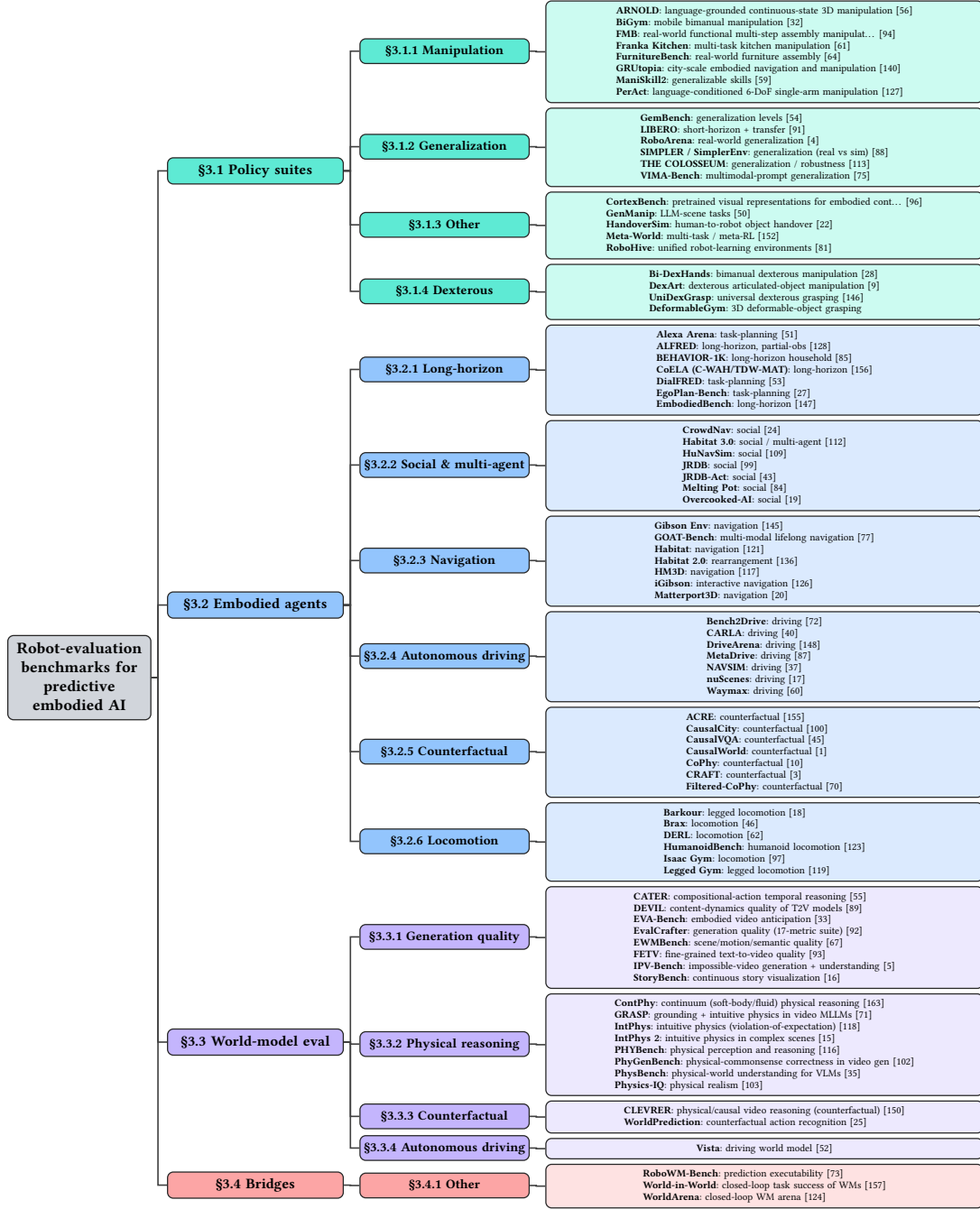


Fig. 7. **A taxonomy of robot-evaluation benchmarks for predictive embodied intelligence** (86 representative benchmarks of the 160 in Table 10). Each sub-family is one cell listing its benchmarks, each with a short statement of what it measures. Benchmarks are grouped by evaluation lane (§3.1–§3.4) and then by the capability they stress. The lanes run from closed-loop task-success suites to the *bridge* frontier (shaded), the only lane that turns prediction into executed action, which is the survey’s novelty axis. Full per-benchmark detail: Tables 3, 10.

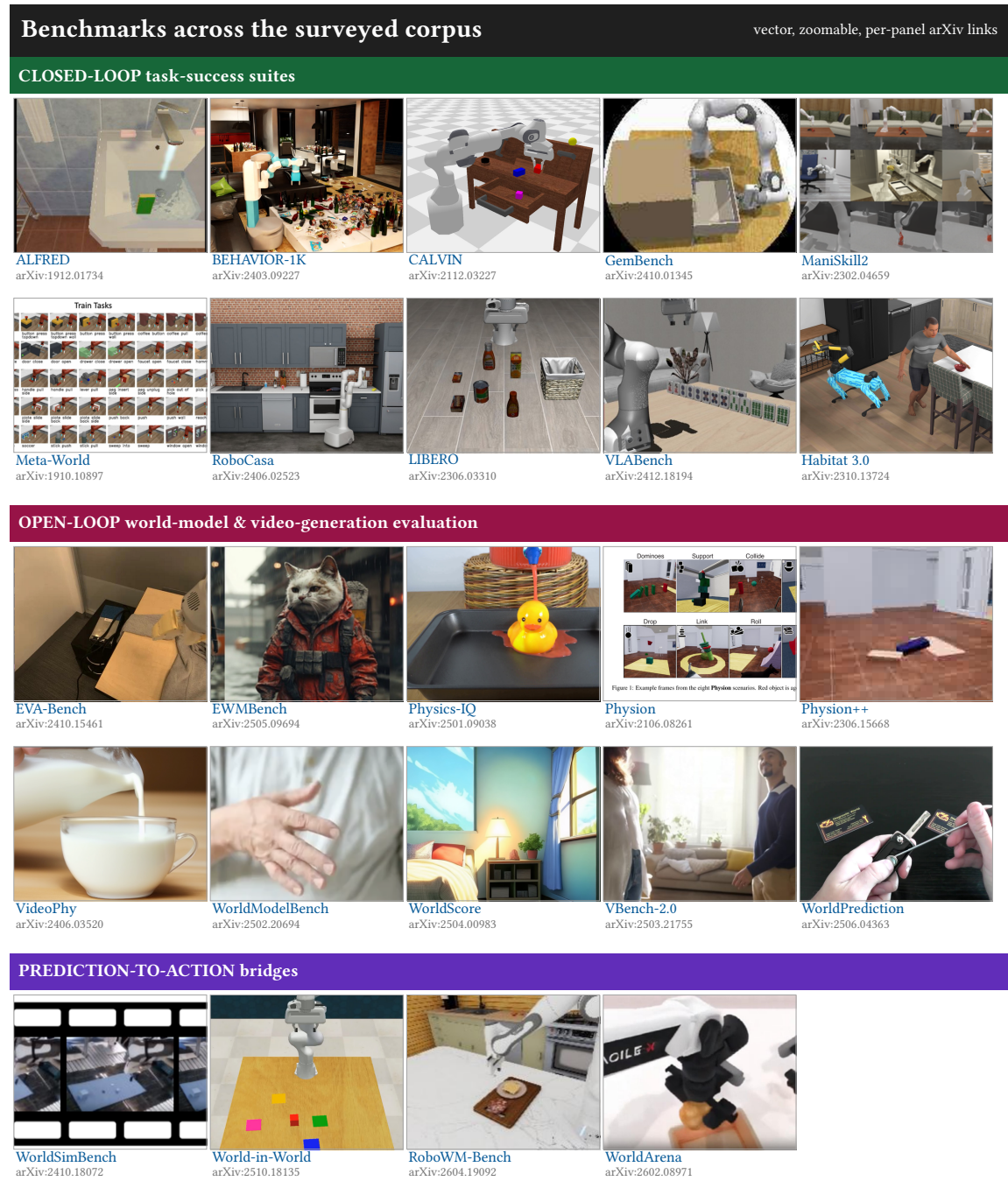


Fig. 8. **Benchmarks across the surveyed corpus.** Representative subject figures from the catalogue, grouped by evaluation lane rather than embodiment: *closed-loop* task-success suites, *open-loop* world-model and video-generation evaluation, and the *prediction-to-action* bridges. Each panel is a live hyperlink to the source paper on arXiv and is annotated with its arXiv id (per-panel figure provenance in Table 11); the figure is vector with photographs embedded at full resolution, so it can be zoomed without pixelation. Only figures that depict the benchmark’s task or scene are shown; results and leaderboard figures appear in Figure 9. Benchmarks with no locatable subject figure are omitted, not fabricated. Images © their respective authors, reproduced for scholarly review.

a world-model policy beats a direct one. Table 5 audits, capability by capability, which benchmarks come closest to building such a contrast, and finds the lane almost entirely agnostic.

Table 5. **Capability deep-dive:** the benchmarks that probe each of the β capability cuts, and whether each builds a native VLA-vs-world-model contrast. Native contrasts cluster in the hidden-state, theory-of-mind and counterfactual *reasoning* cuts (LIBERO-Mem, ComPhy, AGENT, BIB, ACRE, CoPhy, CausalWorld); the long-horizon and social *task-success* cuts are largely model-agnostic (ALFWorld’s abstract-to-grounded transfer is the lone partial). **Eval:** *open/closed*. **VLA vs WM:** ✓ native contrast, ~ partial, -- model-agnostic; each cut’s header tallies the fraction of its benchmarks that build any (native or partial) contrast.

Benchmark	Year	Eval	What it probes	VLA vs WM
Hidden-state / memory / occlusion				3/4
LIBERO-Mem [36]	2025	closed	occluded object state carried across time	✓
TEACH [107]	2021	closed	partner belief and history via dialog	--
ComPhy [29]	2022	open	hidden mass and charge inferred from collisions	✓
Bongard-HOI [74]	2022	open	few-shot hidden-concept induction	~
Theory-of-mind				3/4
AGENT [130]	2021	open	expectations over an agent’s goals and actions	✓
BIB [48]	2021	open	infant-style goal inference from behaviour	✓
PHASE [106]	2021	closed	social relations and goals inferred from motion	~
SocialAI [78]	2021	closed	language-based social interaction with agents	--
Counterfactual / physical inference				3/6
WorldPrediction [25]	2025	open	discriminate correct vs counterfactual action	--
Physion [12]	2021	open	predict a physical outcome from vision	--
Physion++ [138]	2023	open	infer latent physical properties online	--
ACRE [155]	2021	open	abstract causal reasoning over blinket tests	✓
CoPhy [10]	2019	open	counterfactual outcome under a changed cause	✓
CausalWorld [1]	2020	closed	causal do-interventions on task variables	✓
Long-horizon / compositional				1/6
CALVIN [101]	2021	closed	chain several language subtasks in a row	--
BEHAVIOR-1K [85]	2024	closed	1000 long-horizon household activities	--
ALFRED [128]	2019	closed	instruction to action under partial view	--
ALFWorld [129]	2020	closed	abstract-text to grounded-task transfer	~
EmbodiedBench [147]	2025	closed	broad long-horizon embodied planning	--
PARTNR [21]	2024	closed	human-robot collaborative household tasks	--
Social / multi-agent				0/5
Habitat 3.0 [112]	2023	closed	human-robot co-habitation and hand-off	--
SocNavBench [14]	2021	closed	socially compliant navigation	--
Melting Pot [84]	2021	closed	mixed-motive multi-agent generalization	--
SMAC [120]	2019	closed	cooperative multi-agent micro-control	--
CrowdNav [24]	2018	closed	socially-aware navigation among pedestrians	--

3.2 Embodied agents

The embodied-agent lane is the largest (85 benchmarks) and the most diverse. It spans *long-horizon* household and planning tasks (§3.2.1: ALFRED [128], BEHAVIOR-1K [85], EmbodiedBench [147]), *social and multi-agent* settings (§3.2.2: Habitat 3.0 [112], Overcooked-AI [19], Melting Pot [84]), *navigation* (§3.2.3: Habitat [121], HM3D [117], GOAT-Bench [77]), *autonomous driving* (§3.2.4: CARLA [40], Bench2Drive [72], Waymax [60]), *counterfactual* reasoning (§3.2.5:

CausalWorld [1], CoPhy [10], ACRE [155]), and *locomotion* (§3.2.6: HumanoidBench [123], Barkour [18], Isaac Gym [97]). Two observations matter for the survey’s question. First, like the policy lane, these suites are overwhelmingly model-agnostic. Second, the counterfactual sub-family, which is where predictive world modelling should most plausibly help, is the thinnest: it is populated by causal-reasoning tasks but almost never run as a closed-loop VLA-versus-world-model contrast.

3.3 World-model evaluation

The world-model-evaluation lane (34 benchmarks) scores prediction and generation directly, with no action taken. Its sub-families are *generation quality* (§3.3.1: VBench [68], EvalCrafter [92], EWMBench [67], EVA-Bench [33]), *physical reasoning* (§3.3.2: IntPhys [118], PhysBench [35], PhyGenBench [102], Physics-IQ [103]), *counterfactual* video reasoning (§3.3.3: CLEVRER [150], WorldPrediction [25]), and a driving world model (§3.3.4: Vista [52]). Table 6 is a deep-dive over all 34: it records what each scores (generation, question-answering, or prediction), the signal it uses (automatic metric, accuracy, violation-of-expectation, or human rating), and whether it touches physical grounding. The lane’s defining limitation is uniform: every benchmark scores open-loop, and none executes what it predicts, so a high generation score certifies visual quality but says nothing about task success.

Table 6. **World-model-evaluation deep-dive:** the 34 benchmarks of the world-model-evaluation lane, by sub-type – the P3 counterpart of the skill-building branch deep-dive. **Mode:** *Gen* evaluates generated video, *QA* video question-answering, *Pred* physical prediction. **Signal:** *Auto* computed/learned metric, *Acc* accuracy vs ground truth, *VOE* violation-of-expectation, *Human* human study. **Physical:** targets physical-law understanding, *✓* core, *~* adjacent, *✗* general. Generation-quality suites dominate and are physics-agnostic; only the physical-reasoning and physical-commonsense sub-types test physical law, and every lane benchmark scores open-loop, never in a closed loop.

Benchmark	Year	Mode	What it scores	Signal	Physical
Generation quality (multi-dimensional)					
EvalCrafter [92]	2023	Gen	17-metric text-to-video quality suite	Auto	✗
FETV [93]	2023	Gen	fine-grained text-to-video quality	Auto	✗
VBench [68]	2023	Gen	16 disentangled generation dimensions	Auto	✗
StoryBench [16]	2023	Gen	continuous story-visualisation quality	Auto	✗
DEVIL [89]	2024	Gen	content-dynamics quality of T2V	Auto	✗
T2V-CompBench [134]	2024	Gen	compositional text-to-video quality	Auto	✗
T2VScore [144]	2024	Gen	learned alignment + quality metric	Auto	✗
TC-Bench [44]	2024	Gen	temporal compositionality in generation	Auto	✗
VideoScore [63]	2024	Gen	learned automatic quality metric	Auto	✗
EWMBench [67]	2025	Gen	scene / motion / semantic quality	Auto	✗
VBench-2.0 [161]	2025	Gen	generation faithfulness	Auto	✗
Intuitive & physical reasoning					
IntPhys [118]	2018	Pred	intuitive physics from video	VOE	✓
Physion [12]	2021	Pred	physical outcome prediction from vision	Acc	✓
GRASP [71]	2023	QA	intuitive physics in video MLLMs	Acc	✓
Physion++ [138]	2023	Pred	prediction with online property inference	Acc	✓
ContPhy [163]	2024	QA	soft-body / fluid physical reasoning	Acc	✓
Physics-IQ [103]	2025	Gen	physical realism of generated video	Auto	✓
PhysBench [35]	2025	QA	physical-world understanding for VLMs	Acc	✓
PHYBench [116]	2025	QA	physical perception and reasoning	Acc	✓
IntPhys 2 [15]	2025	Pred	intuitive physics in complex scenes	VOE	✓
Physical commonsense in generated video					
PhyGenBench [102]	2024	Gen	physical-commonsense correctness in gen	Auto	✓

Table 6 (continued)

Benchmark	Year	Mode	What it scores	Signal	Physical
VideoPhy [7]	2024	Gen	physical commonsense in generated video	Human	✓
VideoPhy-2 [8]	2025	Gen	action-centric physical commonsense	Human	✓
PhyWorldBench [58]	2025	Gen	physical realism in text-to-video	Human	✓
IPV-Bench [5]	2025	Gen	impossible-video generation + understanding	Acc	✓
Counterfactual / causal / temporal					
CATER [55]	2019	QA	compositional-action temporal reasoning	Acc	✗
CLEVRER [150]	2019	QA	causal / counterfactual video reasoning	Acc	~
WorldPrediction [25]	2025	QA	counterfactual action recognition	Acc	✗
Video-language alignment / hallucination					
VideoCon [6]	2023	QA	robust video-language alignment	Auto	✗
VideoHalluciner [141]	2024	QA	hallucination in video-language models	Acc	✗
World-model prediction & world generation					
Vista [52]	2024	Gen	driving world-model generation	Auto	~
EVA-Bench [33]	2024	Pred	embodied video anticipation	Auto	~
WorldModelBench [86]	2025	Gen	video world-model prediction quality	Human	~
WorldScore [41]	2025	Gen	controllable world-generation quality	Auto	✗

3.4 Prediction-to-action bridges

The bridge lane is the frontier, and it is tiny: four benchmarks. RoboWM-Bench [73] tests whether a world model’s predictions are *executable*; World-in-World [157] and WorldArena [124] place world models in a closed loop and score task success rather than rollout realism; WorldSimBench [115] scores generative models both as renderers and as controllable simulators. These are the only benchmarks that carry prediction all the way into executed action, and they are the shaded frontier of Figure 7. Even so, none of the four yet reports a per-capability, head-to-head comparison between a direct VLA policy and a world-model policy under one protocol. The bridges prove the loop can be closed; they do not yet close it in a way that isolates the advantage of prediction. That is the gap Section 4 formalises.

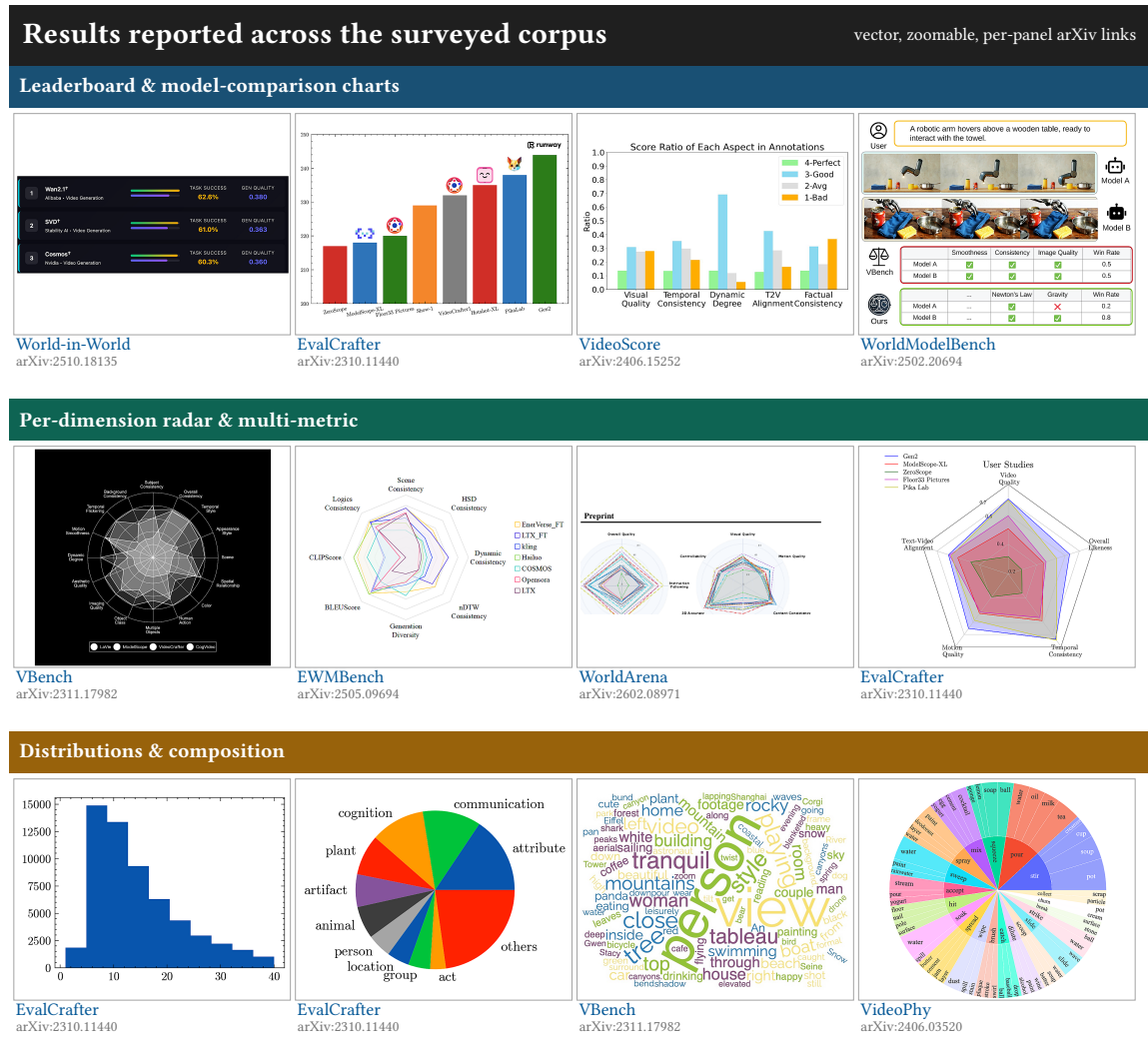


Fig. 9. Results reported across the surveyed corpus. Representative results figures from the catalogue, grouped by type of evidence: *leaderboard and model-comparison charts*, *per-dimension radar and multi-metric plots*, and *distributions and composition*. Each panel is a live hyperlink to the source paper on arXiv and is annotated with its arXiv id (per-panel provenance in Table 12); the figure is vector with plots embedded at full resolution. Only results/leaderboard/metric figures are shown; task and scene figures appear in Figure 8. As a benchmark survey the reported evidence is dominated by rankings, radars and distributions rather than training curves, and is sparser than the subject figures, so this gallery is smaller; benchmarks that publish no results figure are omitted, not fabricated. A benchmark may appear more than once when it reports several evidence types. Images © their respective authors, reproduced for scholarly review.

4 DOES PREDICTION HELP ACTING? THE CONTRAST GAP

The survey’s question can be stated precisely. Let a *direct policy* map observations to actions, and let a *world-model policy* first predict future observations and then plan an action against that prediction. The advantage of prediction is the closed-loop task-success *gain* of the world-model policy over a matched direct policy, and it is only meaningful when reported per capability, since prediction should help more where hidden state, occlusion, or long horizons make foresight valuable. Figure 10 draws the evaluation loop this definition requires: a task forks into a direct-VLA branch and a world-model branch, both run in one closed loop with state feedback, both scored by the same task-success measure, and the difference read off per capability.

Today’s benchmarks break this loop in one of two ways, visible as the two poles of Figure 1. Model-agnostic policy and embodied suites (Sections 3.1, 3.2) close the loop but run a single branch, so there is no contrast to read. World-model-evaluation suites (Section 3.3) score the prediction branch open-loop and never execute it, so a high score certifies generation quality, not task success. The prediction-to-action bridges (Section 3.4) close the loop for a world-model policy but do not yet run the matched VLA branch per capability. No lane completes the diagram.

4.1 “World model” is three things, none of them sufficient alone

Part of why the contrast is hard to build is that the term “world model” is overloaded. Table 7 separates two axes the literature routinely conflates. Along a *representation* axis a world model may be a latent-dynamics predictor (LWM; DreamerV3, TD-MPC2), an action-conditioned future-frame predictor (WAM; iVideoGPT, GR-2), or a generative video model (WFM; Sora, Cosmos, Genie). Along a *role* axis it may be a neural simulator that trains or evaluates a policy, or the object that a benchmark scores. The crucial fact is that no single sense is simultaneously action-conditioned, run in a closed loop, and readily comparable to a VLA policy under one protocol: latent and action-conditioned models act but are rarely scored by a standard generation metric, while generative models carry the metrics but are rarely closed into a loop. A benchmark that contrasts a world-model policy with a VLA policy therefore has to assemble properties that no existing sense of the term supplies at once.

4.2 Why each lane leaves the contrast unbuilt

Table 8 compares the four lanes on six evaluation axes and makes the pattern lane-by-lane. Policy and embodied suites score the right quantity, closed-loop success, on the right capabilities, but hold the model family fixed and so never contrast it. World-model-evaluation suites vary the model family richly but score the wrong quantity for our question, open-loop prediction quality, and never close the loop. The bridges are the only lane that scores closed-loop success of a world model, but they are four benchmarks and none slices a VLA head-to-head by capability. The result is a coverage hole precisely where the survey’s question lives: the intersection of closed-loop scoring, per-capability slicing, and a world-model-versus-VLA family contrast.

4.3 Measuring the advantage: assessed, not adopted

Closing the loop is necessary but not sufficient; the gain also needs an instrument. Proposals exist, including behavioural advantage-of-prediction scores such as ACTION-ATLAS’s World Advantage Score, but they are put forward in position papers [153] rather than established across a benchmark suite. We therefore *assess* such instruments as candidates rather than adopting any one as ground truth, and Section 5 specifies how an advantage-of-prediction quantity should be reported: as a per-capability distribution or curve against a matched baseline, never as a single headline number. The

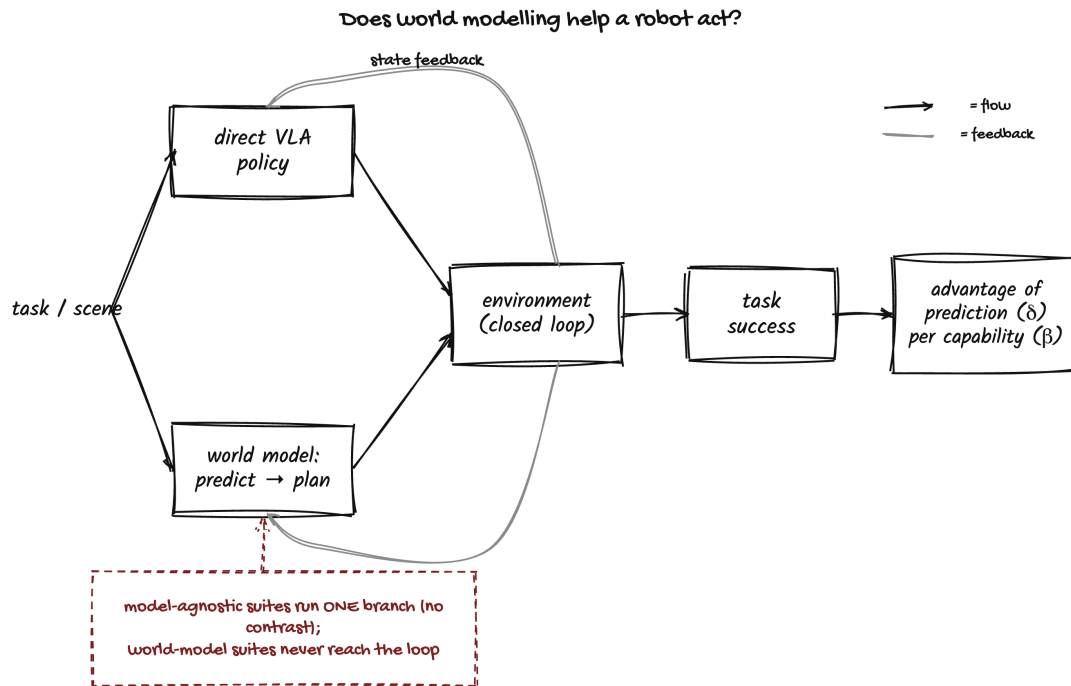


Fig. 10. **The evaluation loop that isolates the advantage of prediction.** A benchmark that can answer “does world modelling help a robot act” must run a direct Vision-Language-Action policy *and* a world-model policy (predict then plan) under one closed loop, then report the closed-loop gain (δ) sliced by capability (β). Today’s suites break this loop in one of two ways: model-agnostic policy suites host a single policy and never build the contrast, while world-model suites score the prediction open-loop and never execute it (§4).

empirical scale of the gap is worth restating: of 160 benchmarks, 138 are model-agnostic, 11 build a partial contrast, and 11 build an explicit one; counterfactual capability, where foresight should matter most, is almost entirely unmeasured; and only four benchmarks reach executed action at all.

Sense of model	“world model”	What it is	Examples	Act. cond.	Loop	Metric	VLA-cmp.
<i>As a representation (what the model is)</i>							
latent WM (LWM)		latent-dynamics predictor $z_{t+1} = f(z_t, a_t)$	DreamerV3, TD-MPC2	✓	✓	~	~
action-cond. (WAM)	WM	action-conditioned future-frame predictor	iVideoGPT, GR-2	✓	~	~	~
generative WM (WFM)		text / image-to-video generator	Sora, Cosmos, Genie	~	✗	✓	✗
<i>As a role (what the model is used for)</i>							
neural simulator		learned environment for policy training / eval	UniSim	✓	✓	~	~
evaluation target		benchmarks that score world models	WorldSimBench, World-ModelBench, EWMBench	--	~	✓	✗

Table 7. The term “world model” spans two orthogonal axes the literature routinely conflates: a *representation* axis (what the model is: latent-state dynamics, action-conditioned video, or generative video, a soft boundary since action-conditioned generators straddle WAM and WFM) and a *role* axis (what it is used for: a neural simulator, or the object a benchmark scores). Columns record whether a sense is **action-conditioned**, is used in a closed **loop**, carries a standard evaluation **metric**, and is readily **VLA-comparable** under one protocol (✓ yes, ~ partial, ✗ no, -- n/a). No single sense is action-conditioned, closed-loop, and VLA-comparable at once, which is why a benchmark that contrasts a world-model policy with a VLA policy per capability (§4) is still missing. Evaluation benchmarks score the world model as an *object*; they are not world models acting as judges.

	Policy suites [91]	Embodied nav / social [112]	World-model eval [86]	Bridges [157]
Loop closure	✓ closed-loop; steps the environment [101]	✓ closed-loop; long-horizon rollouts [85]	✗ open-loop; scores generation, no action [103]	~ converts a prediction into an executed action [115]
Task horizon	~ short reactive skills; long by chaining [91]	✓ long-horizon household and dialog [107]	~ short clips; temporal but not task-level [67]	~ a single executed rollout [73]
Capability breadth	~ manipulation + transfer; hidden-state rare [113]	✓ navigation, social, some hidden-state [38]	~ physical / temporal; counterfactual barely [25]	✗ narrow; executability only [124]
Model-family contrast	✗ model-agnostic host of any policy [4]	~ memory-vs-Markovian in one suite [36]	✗ world-model-only scores [86]	~ WM as policy substrate; no VLA pole [157]
Reproducibility	~ sim; some real-vs-sim transfer [88]	~ sim-heavy; a sim-to-real gap [38]	✓ fixed offline sets; deterministic scoring [161]	✗ new, still-unstandardised protocols [124]
Structural blind spot	hosts a world-model policy but never builds the VLA contrast	capability cuts exist, but no per-capability family ablation	visual quality is not task success; no behavioural utility	reaches action, but omits the per-capability VLA head-to-head

Table 8. **Evaluation-lane comparison matrix:** the four benchmark lanes on six evaluation axes (loop closure, task horizon, capability breadth, model-family contrast, reproducibility, and characteristic blind spot). Marks grade each lane on the axis (✓ favourable, ~ mixed, ✗ weak), with an anchoring benchmark per cell. No lane closes the loop, slices capability, *and* builds a native VLA-vs-world-model contrast at once; the bridge lane comes closest yet still omits the per-capability VLA head-to-head, which is the empty cell this survey names (§4).

5 AN AGENDA FOR ADVANTAGE-AWARE EVALUATION

If the gap is that no benchmark is built to ask whether prediction helps acting, the remedy is a small, measurable protocol that any benchmark could adopt. Table 9 sets out four quantities, each with a definition, a concrete testbed, and the axis it stresses, and Figure 11 sketches how they compose. The figure shows target shapes, not measured results: it is a protocol, not a leaderboard.

The first quantity is an *advantage-of-prediction curve*: the closed-loop success gain of a world-model policy over a matched direct-VLA baseline, reported per capability as a curve rather than a single endpoint, instantiated over LIBERO [91] and CALVIN [101] run under one harness. The second is *counterfactual accuracy*: task success on episodes

Metric	Definition	Instantiation (testbed)	Axis stressed
Advantage-of-prediction curve	closed-loop success <i>gain</i> of a predictive policy over a matched direct-VLA baseline, reported per capability as a curve, not a single endpoint	a capability-sliced protocol over LIBERO [91] and CALVIN [101] that runs a world-model policy and a VLA policy under one harness	γ family, δ metric
Counterfactual accuracy	task success on episodes that require inference over unobserved or hypothetical dynamics (occlusion, object permanence, “what if”)	WorldPrediction [25] plus occlusion / memory splits in the style of LIBERO-Mem [36]	β capability
Prediction-to-action fidelity gap	the difference between a world model’s open-loop generation quality and its closed-loop task success (“visual quality is not task success”)	bridge protocols such as WorldSimBench [115] and World-in-World [157]	α mode
Per-capability contrast coverage	the fraction of capabilities for which a benchmark actually runs a VLA-vs-world-model head-to-head under a single protocol	audits of the landscape (Table 10) against a declared contrast schema	γ family, ε catalog

Table 9. **An actionable protocol for the future agenda:** four measurable quantities, their definitions, and concrete testbeds. The first anchors the advantage-of-prediction question (δ) on LIBERO / CALVIN-style suites; the second targets the near-unmeasured counterfactual capability; the third quantifies the prediction-to-action gap the bridges expose; the fourth audits per-capability contrast coverage across the landscape. Each is reported as a distribution or curve, never a single headline number.

that require reasoning over unobserved or hypothetical dynamics, built from WorldPrediction [25] together with occlusion and memory splits in the style of LIBERO-Mem [36]; this targets the capability the corpus most neglects. The third is the *prediction-to-action fidelity gap*: the difference between a world model’s open-loop generation quality and its closed-loop task success, measured on bridge protocols such as WorldSimBench [115] and World-in-World [157], which quantifies directly the “visual quality is not task success” warning of Section 4. The fourth is *per-capability contrast coverage*: the fraction of capabilities for which a benchmark actually runs a VLA-versus-world-model head-to-head, computed by auditing the landscape (Table 10) against a declared contrast schema. Each is reported as a distribution or a curve, never as one number, so that “prediction helps” can be replaced by “prediction helps *here*, by *this much*”.

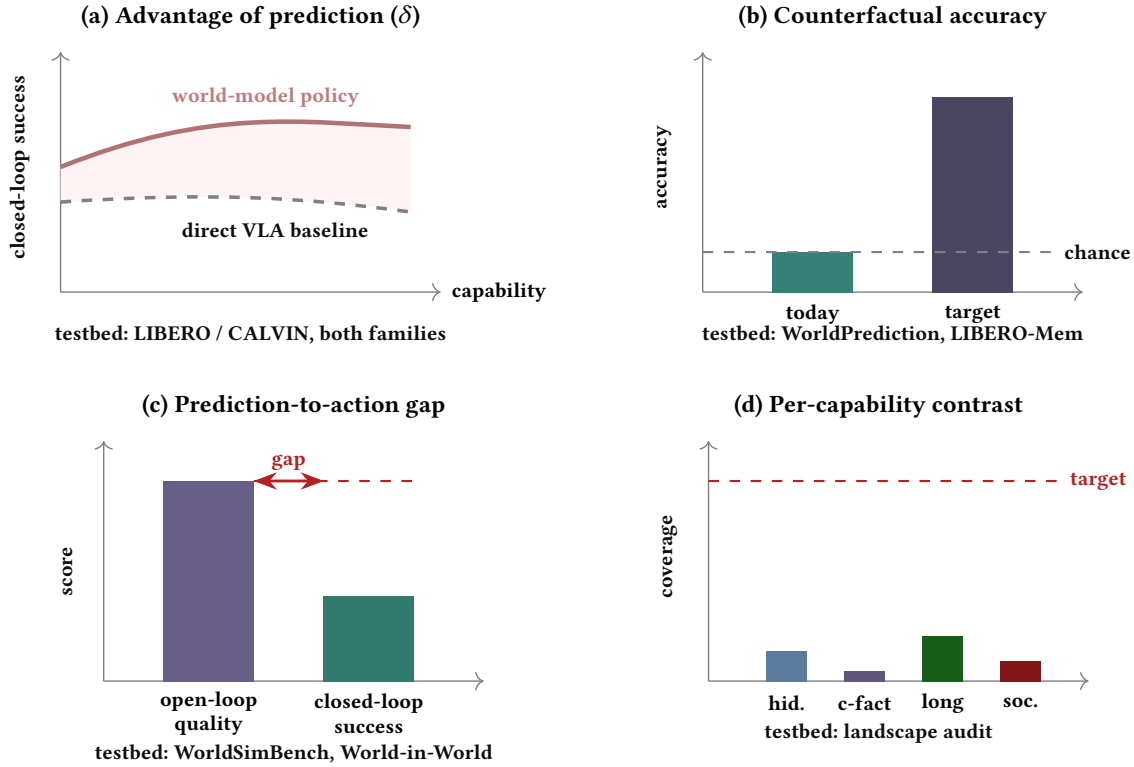


Fig. 11. **An actionable evaluation protocol for the future agenda** (Table 9). The panels are *schematic*: they show the axes and the *target* shape of each measurement, not measured results. (a) a world-model policy should beat a matched direct-VLA baseline in closed-loop success, and the per-capability gap is the advantage of prediction (δ), on LIBERO / CALVIN-style suites run with both families; (b) counterfactual accuracy is near chance today and should rise, on WorldPrediction and occlusion splits; (c) a world model’s open-loop quality far exceeds its closed-loop task success, and that gap must be reported, on bridge suites; (d) the fraction of capabilities with a native VLA-vs-world-model contrast is far below target across the landscape.

6 LIMITATIONS

Three limits bound the survey’s claims. First, its unit is the benchmark, not the model: the corpus maps how the field *measures* predictive embodied intelligence and deliberately says nothing about which world models or VLA policies are strongest. A reader looking for a model ranking will not find one here, by design. Second, the contrast axis rests on an operational definition (Table 2): a benchmark counts as building a VLA-versus-world-model contrast only when both branches are run under one protocol. A more permissive reading would move some benchmarks from model-agnostic to partial, though the qualitative picture, that explicit contrasts are rare and counterfactual coverage is thin, is robust to the exact threshold. Third, coverage is biased towards recent, arXiv-indexed, English-language work reachable by web search; the funnel in Figure 3 records only logged stages, and the two galleries include only benchmarks whose figures could be located and provenance-checked, so absence from a gallery is not evidence of absence of results. Finally, the advantage-of-prediction instruments discussed in Section 4 are assessed as candidates rather than adopted; we do not rank benchmarks by any metric that the field has not yet validated.

7 CONCLUSION

Do world models make better robots? On the evidence of the benchmarks the field has built, the honest answer is that we cannot yet tell. The question asks for a closed-loop, per-capability comparison between a world-model policy and a direct VLA policy, and our catalogue of 160 web-verified benchmarks shows that almost none is built to run it: 138 of 160 are model-agnostic, only 11 build an explicit contrast, counterfactual capability is nearly unmeasured, and only four benchmarks carry prediction into executed action. The obstacle is methodological. World-model suites score prediction without executing it, task-success suites execute a single policy without contrasting families, and the term “world model” itself names three different objects, none of which is action-conditioned, closed-loop, and VLA-comparable at once.

This survey’s contribution is to make that gap precise and measurable. We give an operational taxonomy of the landscape across four evaluation lanes, a coverage comparison showing that no prior survey crosses capability with model family, an evaluation loop that isolates the advantage of prediction, and a four-metric protocol, anchored on named testbeds, that any benchmark could adopt to report where and by how much prediction helps. The next benchmark that matters will not be the one with the most tasks or the most realistic renders; it will be the one that runs both branches in one loop and reads off the difference. Building it is the empty cell this survey names.

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A THE FULL BENCHMARK LANDSCAPE

The core taxonomy (Figure 7) shows 86 representative benchmarks, those that are cleanly axis-placeable, distinct, and fully characterisable from their papers. For completeness and reproducibility, Table 10 lists the full catalogue of 160 web-verified benchmarks, grouped by the same four evaluation lanes and sub-grouped by capability, and compared on year, capability, evaluation mode, and whether the benchmark builds a native VLA-versus-world-model contrast. Every entry is a real, individually citeable benchmark; the column tallies are the source of the counts reported throughout the survey and of the distributions in Figure 5.

Table 10. **The robot-evaluation benchmark landscape: 160 web-verified benchmarks, compared on four axes.** Grouped into four evaluation lanes and sub-grouped by capability. Columns: **Year**; **Capability** (what the benchmark stresses); **Eval** (evaluation mode: open-loop prediction/generation quality, closed-loop task success, or a prediction-to-action bridge); and **VLA vs WM** (does the benchmark build a native direct-policy vs world-model contrast: ✓ explicit, ~ partial, -- model-agnostic). Every entry is a real, individually citeable benchmark.

Benchmark	Year	Capability	Eval	VLA vs WM
Policy / manipulation suites (37)				
<i>Manipulation</i>				
ARNOLD [56]	2023	language-grounded continuous-state 3D manipulation	closed	--
BiGym [32]	2024	mobile bimanual manipulation	closed	--
FMB [94]	2024	real-world functional multi-step assembly manipulation	closed	--
Franka Kitchen [61]	2019	multi-task kitchen manipulation	closed	--
FurnitureBench [64]	2023	real-world furniture assembly	closed	--
GRUtopia [140]	2024	city-scale embodied navigation and manipulation	closed	--
KitchenShift	2021	kitchen distribution-shift generalization	closed	--
ManiSkill2 [59]	2023	generalizable skills	closed	--
PerAct [127]	2022	language-conditioned 6-DoF single-arm manipulation	closed	--
PerAct2 [57]	2024	bimanual language-conditioned manipulation	closed	--
Ravens [154]	2020	vision-based pick-and-place rearrangement	closed	--
RGB-Stacking [83]	2021	vision-based stacking of diverse shapes	closed	--
RLBench [69]	2020	manipulation breadth	closed	--
RoboAgent [13]	2023	multi-task manipulation with RoboSet	closed	--
robomimic [98]	2021	single-arm manipulation from human demonstrations	closed	--
robosuite [164]	2020	modular simulated single-arm manipulation tasks	closed	--
RoboTwin [104]	2024	dual-arm bimanual manipulation	closed	--
VLMBench [162]	2022	vision-and-language compositional manipulation	closed	--
<i>Generalization & robustness</i>				
GemBench [54]	2024	generalization levels	closed	--
LIBERO [91]	2023	short-horizon + transfer	closed	--
RoboArena [4]	2025	real-world generalization	closed	--
SIMPLER / SimplerEnv [88]	2024	generalization (real vs sim)	closed	--
THE COLOSSEUM [113]	2024	generalization / robustness	closed	--
VIMA-Bench [75]	2023	multimodal-prompt generalization	closed	--
<i>Other</i>				
CortexBench [96]	2023	pretrained visual representations for embodied control	closed	--
GenManip [50]	2025	LLM-scene tasks	closed	~
HandoverSim [22]	2022	human-to-robot object handover	closed	--
Meta-World [152]	2020	multi-task / meta-RL	closed	--
RoboHive [81]	2023	unified robot-learning environments	closed	--
<i>Dexterous manipulation</i>				
Bi-DexHands [28]	2022	bimanual dexterous manipulation	closed	--
DeformableGym	2023	3D deformable-object grasping	closed	--
DexArt [9]	2023	dexterous articulated-object manipulation	closed	--
UniDexGrasp [146]	2023	universal dexterous grasping	closed	--

Table 10 (continued)

Benchmark	Year	Capability	Eval	VLA vs WM
<i>Long-horizon & planning</i>				
CALVIN [101]	2021	long-horizon language	closed	--
RoboCasa [105]	2024	long-horizon (data scaling)	closed	--
VLABench [160]	2025	long-horizon reasoning	closed	--
<i>Deformable-object manipulation</i>				
SoftGym [90]	2020	deformable-object manipulation	closed	--
Embodied navigation, planning & social (85)				
<i>Long-horizon & planning</i>				
Alexa Arena [51]	2023	task-planning	closed	--
ALFRED [128]	2019	long-horizon, partial-obs	closed	--
BEHAVIOR-1K [85]	2024	long-horizon household	closed	--
CoELA (C-WAH/TDW-MAT) [156]	2023	long-horizon	closed	--
DialFRED [53]	2022	task-planning	closed	--
EgoPlan-Bench [27]	2023	task-planning	closed	--
EmbodiedBench [147]	2025	long-horizon	closed	--
EmbodiedCity [49]	2024	long-horizon	closed	--
EmbodiedEval [30]	2025	long-horizon	closed	--
HomeRobot OVMM [149]	2023	long-horizon	closed	--
LEGENT [31]	2024	long-horizon	closed	--
LH-VLN [131]	2024	long-horizon	closed	--
LoHoRavens [159]	2023	long-horizon	closed	--
LoTa-Bench [34]	2024	task-planning	closed	--
MFE-ETP [158]	2024	task-planning	closed	--
PARTNR [21]	2024	task-planning	closed	--
PlanBench [139]	2022	task-planning	closed	--
SafeAgentBench [151]	2024	task-planning	closed	--
SMART-LLM [76]	2023	task-planning	closed	--
TDW-Transport [47]	2021	task-planning	closed	--
TidyBot [143]	2023	task-planning	closed	--
VirtualHome [110]	2018	task-planning	closed	--
Watch-And-Help [111]	2020	long-horizon	closed	--
<i>Social & multi-agent</i>				
CrowdNav [24]	2018	social	closed	--
Habitat 3.0 [112]	2023	social / multi-agent	closed	--
HuNavSim [109]	2023	social	closed	--
JRDB [99]	2019	social	closed	--
JRDB-Act [43]	2021	social	closed	--
Melting Pot [84]	2021	social	closed	--
Overcooked-AI [19]	2019	social	closed	--
RoboSense [133]	2024	social	closed	--
SEAN [137]	2020	social	closed	--
SMAC [120]	2019	social	closed	--
SocialGym 2.0 [132]	2023	social	closed	--
SocNavBench [14]	2021	social navigation	closed	--
<i>Navigation</i>				
Gibson Env [145]	2018	navigation	closed	--
GOAT-Bench [77]	2024	multi-modal lifelong navigation	closed	--
Habitat [121]	2019	navigation	closed	--
Habitat 2.0 [136]	2021	rearrangement	closed	--
HM3D [117]	2021	navigation	closed	--
iGibson [126]	2020	interactive navigation	closed	--
Matterport3D [20]	2017	navigation	closed	--
ObjectNav [11]	2020	object-goal navigation	closed	--
ProcTHOR [39]	2022	navigation	closed	--

Table 10 (continued)

Benchmark	Year	Capability	Eval	VLA vs WM
RoboTHOR [38]	2020	navigation / sim-to-real	closed	--
SoundSpaces [23]	2020	audio-visual navigation	closed	--
<i>Autonomous driving</i>				
Bench2Drive [72]	2024	driving	closed	--
CARLA [40]	2017	driving	closed	--
DriveArena [148]	2024	driving	closed	✓
MetaDrive [87]	2021	driving	closed	--
NAVSIM [37]	2024	driving	closed	--
nuScenes [17]	2019	driving	closed	--
Waymax [60]	2023	driving	closed	--
Waymo Open [135]	2019	driving	closed	--
<i>Counterfactual reasoning</i>				
ACRE [155]	2021	counterfactual	open	✓
CausalCity [100]	2021	counterfactual	closed	~
CausalVQA [45]	2025	counterfactual	open	✓
CausalWorld [1]	2020	counterfactual	closed	✓
CoPhy [10]	2019	counterfactual	open	✓
CRAFT [3]	2020	counterfactual	open	✓
Filtered-CoPhy [70]	2022	counterfactual	open	✓
<i>Locomotion</i>				
Barkour [18]	2023	legged locomotion	closed	--
Brax [46]	2021	locomotion	closed	--
DERL [62]	2021	locomotion	closed	--
HumanoidBench [123]	2024	humanoid locomotion	closed	--
Isaac Gym [97]	2021	locomotion	closed	--
Legged Gym [119]	2021	legged locomotion	closed	--
<i>Theory of mind</i>				
AGENT [130]	2021	theory-of-mind	open	✓
BIB [48]	2021	theory-of-mind	open	✓
PHASE [106]	2021	theory-of-mind	closed	~
SocialAI [78]	2021	theory-of-mind	closed	--
ToMi	2019	theory-of-mind	closed	~
<i>Vision-language navigation</i>				
R2R [2]	2018	VLN	closed	--
REVERIE [114]	2020	VLN	closed	--
RxR [80]	2020	VLN	closed	--
Touchdown [26]	2019	outdoor VLN	closed	--
VLN-CE [79]	2020	VLN	closed	--
<i>Hidden-state / object permanence</i>				
Bongard-HOI [74]	2022	hidden-state	open	~
ComPhy [29]	2022	hidden-state	open	✓
LIBERO-Mem [36]	2025	hidden-state / occlusion	closed	✓
TEAch [107]	2021	hidden-state (dialog)	closed	--
<i>Embodied question answering</i>				
OpenEQA	2024	EQA	closed	--
RoboVQA [122]	2023	EQA	closed	--
<i>Generalization & robustness</i>				
ALFWorld [129]	2020	abstract-vs-grounded transfer	closed	~
<i>Other</i>				
O-PIAAGETS	2022	object permanence	closed	~
World-model & video-generation evaluation ⁽³⁴⁾				
<i>Generation quality</i>				
CATER [55]	2019	compositional-action temporal reasoning	open	--
DEVIL [89]	2024	content-dynamics quality of T2V models	open	--

Table 10 (continued)

Benchmark	Year	Capability	Eval	VLA vs WM
EVA-Bench [33]	2024	embodied video anticipation	open	--
EvalCrafter [92]	2023	generation quality (17-metric suite)	open	--
EWMBench [67]	2025	scene/motion/semantic quality	open	--
FETV [93]	2023	fine-grained text-to-video quality	open	--
IPV-Bench [5]	2025	impossible-video generation + understanding	open	--
StoryBench [16]	2023	continuous story visualization	open	--
T2V-CompBench [134]	2024	compositional text-to-video quality	open	--
T2VScore [144]	2024	text-to-video metric (alignment + quality)	open	--
TC-Bench [44]	2024	temporal compositionality in video generation	open	--
VBench [68]	2023	generation quality (16 disentangled dimensions)	open	--
VBench-2.0 [161]	2025	generation faithfulness	open	--
VideoCon [6]	2023	robust video-language alignment	open	--
VideoHalluciner [141]	2024	hallucination in video-language models	open	--
VideoScore [63]	2024	learned automatic quality metric	open	--
WorldModelBench [86]	2025	video-WM prediction quality	open	--
WorldScore [41]	2025	world-gen quality	open	--
<i>Physical reasoning</i>				
ContPhy [163]	2024	continuum (soft-body/fluid) physical reasoning	open	--
GRASP [71]	2023	grounding + intuitive physics in video MLLMs	open	--
IntPhys [118]	2018	intuitive physics (violation-of-expectation)	open	--
IntPhys 2 [15]	2025	intuitive physics in complex scenes	open	--
PHYBench [116]	2025	physical perception and reasoning	open	--
PhyGenBench [102]	2024	physical-commonsense correctness in video gen	open	--
PhysBench [35]	2025	physical-world understanding for VLMs	open	--
Physics-IQ [103]	2025	physical realism	open	--
Physion [12]	2021	physical prediction from vision	open	--
Physion++ [138]	2023	physical prediction with online property inference	open	--
PhyWorldBench [58]	2025	physical realism in text-to-video	open	--
VideoPhy [7]	2024	physical commonsense in generated video	open	--
VideoPhy-2 [8]	2025	action-centric physical commonsense	open	--
<i>Counterfactual reasoning</i>				
CLEVRER [150]	2019	physical/causal video reasoning (counterfactual)	open	--
WorldPrediction [25]	2025	counterfactual action recognition	open	--
<i>Autonomous driving</i>				
Vista [52]	2024	driving world model	open	--
Bridges (prediction to action) ⁽⁴⁾				
<i>Other</i>				
RoboWM-Bench [73]	2026	prediction executability	bridge	~
World-in-World [157]	2025	closed-loop task success of WMs	bridge	~
WorldArena [124]	2026	closed-loop WM arena	bridge	~
<i>Generation quality</i>				
WorldSimBench [115]	2024	video-to-action consistency	bridge	~

B PROVENANCE OF THE GALLERIES AND COLLAGE

Every panel in the subject and results galleries (Figures 8, 9) and every tile in the hero collage (Figure 1) is a real figure reproduced from the benchmark’s own paper, hyperlinked to its arXiv source. The tables below record, for each panel and tile, its source benchmark and the verified figure number within that paper; a panel whose figure could not be located was dropped rather than guessed. Table 11 covers the subject gallery, Table 12 the results gallery, and Table 13 the collage.

Benchmark	arXiv	Panel source (subject scene)
Closed-loop task-success suites		
ALFRED	1912.01734	sim task scene (rep. figure, pp.1–3)
BEHAVIOR-1K	2403.09227	household sim scene (rep. figure, pp.1–3)
CALVIN	2112.03227	tabletop manipulation scene (rep. figure, pp.1–3)
GemBench	2410.01345	manipulation scene (rep. figure, pp.1–3)
ManiSkill2	2302.04659	multi-task sim scenes (rep. figure, pp.1–3)
Meta-World	1910.10897	task-suite thumbnail grid (rep. figure, pp.1–3)
RoboCasa	2406.02523	kitchen sim scene (rep. figure, pp.1–3)
LIBERO	2306.03310	manipulation task scene (embedded raster, pp.1–8)
VLABench	2412.18194	mahjong-tile manipulation scene (embedded raster, pp.1–8)
Habitat 3.0	2310.13724	human–robot collaboration scene (embedded raster, pp.1–8)
Open-loop world-model & video-generation evaluation		
EVA-Bench	2410.15461	embodied-anticipation scene (rep. figure, pp.1–3)
EWMBench	2505.09694	generated-video sample (rep. figure, pp.1–3)
Physics-IQ	2501.09038	physical-realism sample (rep. figure, pp.1–3)
Physion	2106.08261	Fig. 1 example-scenario frames (p.3)
Physion++	2306.15668	physical-prediction scene (rep. figure, pp.1–3)
VideoPhy	2406.03520	generated-video sample (rep. figure, pp.1–3)
WorldModelBench	2502.20694	generated-video sample (rep. figure, pp.1–3)
WorldScore	2504.00983	generated world sample (rep. figure, pp.1–3)
VBench-2.0	2503.21755	generated-video sample (embedded raster, pp.1–8)
WorldPrediction	2506.04363	procedural-action video frame (embedded raster, pp.1–8)
Prediction-to-action bridges		
WorldSimBench	2410.18072	video-prediction filmstrip, interactive-eval figure (embedded raster, pp.1–8)
World-in-World	2510.18135	closed-loop manipulation scene (embedded raster, pp.1–8)
RoboWM-Bench	2604.19092	manipulation rollout scene (embedded raster, pp.1–8)
WorldArena	2602.08971	dual-arm manipulation scene (embedded raster, pp.1–8)

Table 11. **Per-panel provenance for Figure 8.** Every panel is a real subject/scene figure cropped from the benchmark’s own arXiv paper and hyperlinked to its source. Exact figure numbers are given where verified from the paper (e.g. Physion Fig. 1); otherwise the teaser-region basis is noted (“rep. figure, pp.1–3” for tiles reused from the collage pool; “embedded raster, pp.1–8” for the nine gap tiles hand-picked from per-paper candidate contact sheets after a subject/results/neither classification pass). No figure number is invented. Images © their respective authors, reproduced for scholarly review.

Benchmark	arXiv	Panel source (results plot)
Leaderboard & model-comparison charts		
World-in-World	2510.18135	task-success / gen-quality leaderboard (embedded raster)
EvalCrafter	2310.11440	overall model-ranking bar chart (embedded raster)
VideoScore	2406.15252	annotation score-ratio grouped bars (embedded raster)
WorldModelBench	2502.20694	VBench-vs-Ours win-rate comparison (embedded raster)
Per-dimension radar & multi-metric		
VBench	2311.17982	16-dimension evaluation radar (embedded raster)
EWMBench	2505.09694	EWMScore 5-dimension radar (embedded raster)
WorldArena	2602.08971	Fig. 1 (p.2): EWMScore ranking bars + evaluation radars
Distributions & composition		
EvalCrafter	2310.11440	prompt-length histogram (embedded raster)
EvalCrafter	2310.11440	meta-type composition pie chart (embedded raster)
VBench	2311.17982	prompt word cloud (embedded raster)

Table 12. **Per-panel provenance for Figure 9.** Every panel is a real results/leaderboard/metric plot cropped from the benchmark’s own arXiv paper and hyperlinked to its source. Exact figure numbers are given where verified from the paper (e.g. WorldArena Fig. 1); otherwise the extraction basis is noted (“embedded raster”). Plots are contain-fitted, not cropped-to-fill, so no bars, axes or radar spokes are cut off. No figure number is invented. A benchmark appears more than once when it reports several evidence types. Images © their respective authors, reproduced for scholarly review.

Benchmark	arXiv	Source	Benchmark	arXiv	Source
Closed-loop (task-success suites)			Open-loop (world-model & video eval)		
LIBERO	2306.03310	Fig. 1 (p.2)	WorldModelBench	2502.20694	rep. figure (pp.1–3)
CALVIN	2112.03227	rep. figure (pp.1–3)	Physics-IQ	2501.09038	rep. figure (pp.1–3)
Meta-World	1910.10897	rep. figure (pp.1–3)	WorldScore	2504.00983	rep. figure (pp.1–3)
ManiSkill2	2302.04659	rep. figure (pp.1–3)	EWMBench	2505.09694	rep. figure (pp.1–3)
RoboCasa	2406.02523	rep. figure (pp.1–3)	EVA-Bench	2410.15461	rep. figure (pp.1–3)
Habitat 3.0	2310.13724	rep. figure (pp.1–3)	VideoPhy	2406.03520	rep. figure (pp.1–3)
ALFRED	1912.01734	rep. figure (pp.1–3)	Physion++	2306.15668	rep. figure (pp.1–3)
GemBench	2410.01345	rep. figure (pp.1–3)	Physion	2106.08261	Fig. 1 (p.3)
VLABench	2412.18194	Fig. 1 Overview (p.1)	EvalCrafter	2310.11440	rep. figure (pp.1–3)

Table 13. **Per-tile provenance for Figure 1.** Every tile is a representative figure cropped from the benchmark’s own arXiv paper (page/figure noted; “rep. figure” marks the paper’s teaser-region figure on pp.1–3), reproduced for scholarly review and hyperlinked to its source. Images © their respective authors.